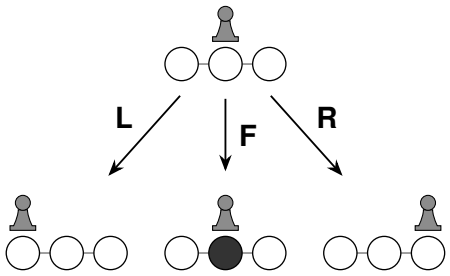
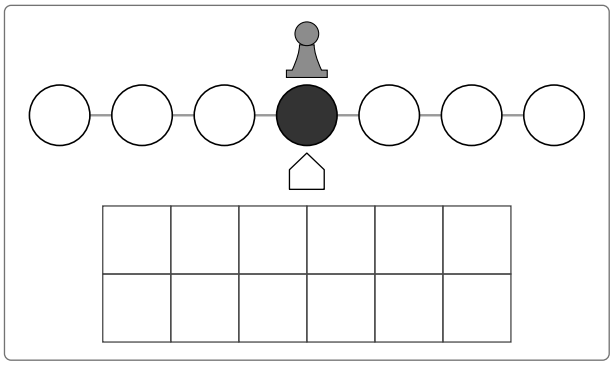
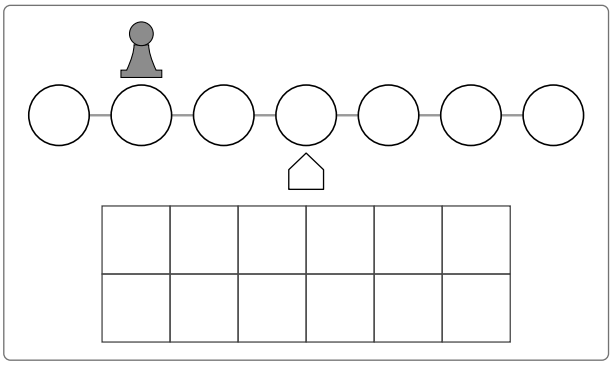
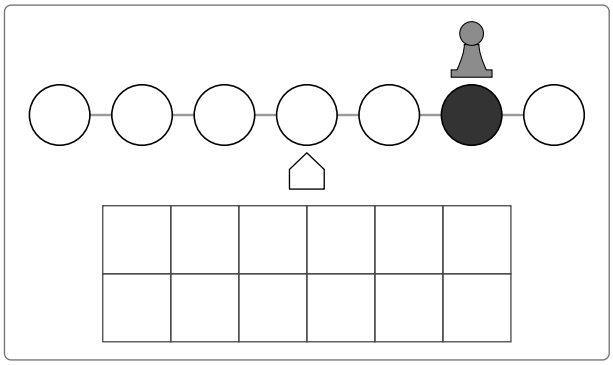
Start every try with all lamps off and the walker at home. A move is one step left, one step right, or a flip of the lamp under the walker. One partner moves the walker. The other points to the lamp under the walker, then flips it. Swap jobs for each new picture. Put one cube on your tower for each move. Color one square under the picture for each cube in your shortest tower.

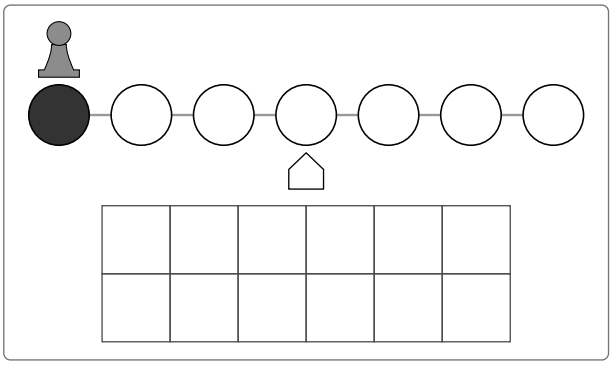


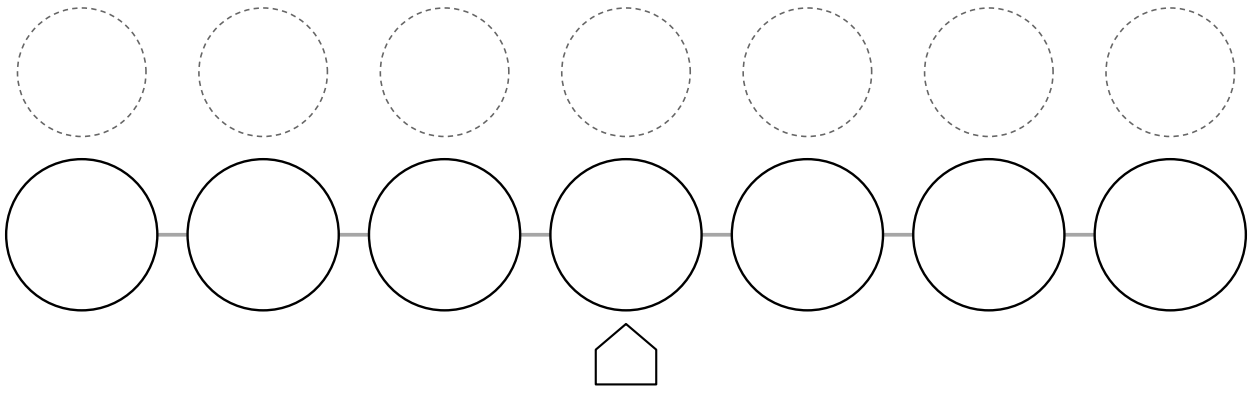
Problem 1: Build each picture with as few moves as you can.



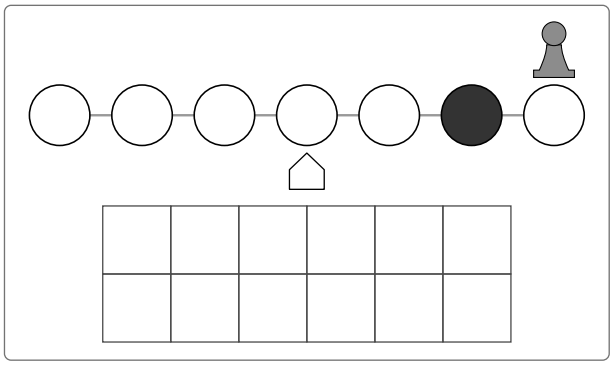


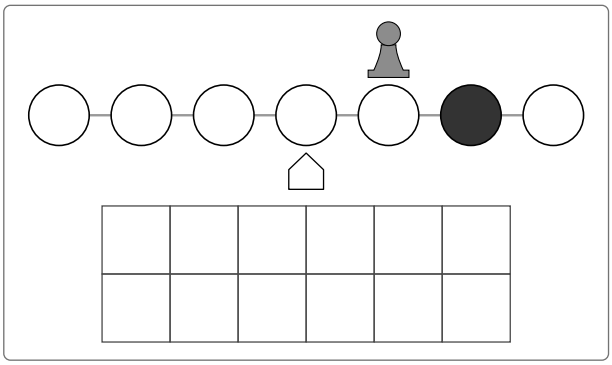


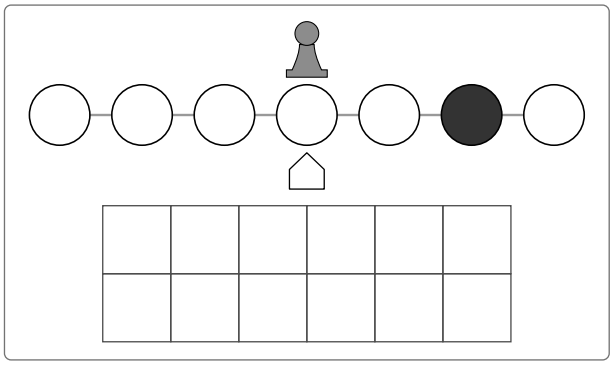


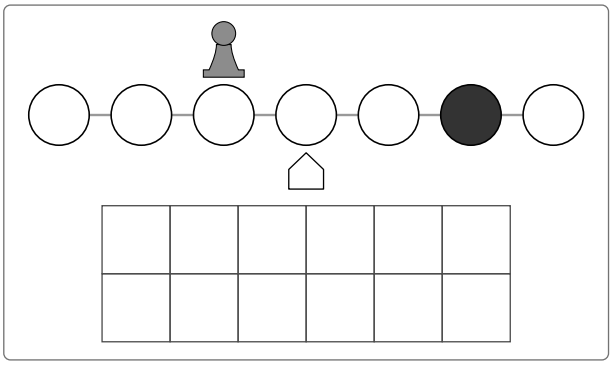


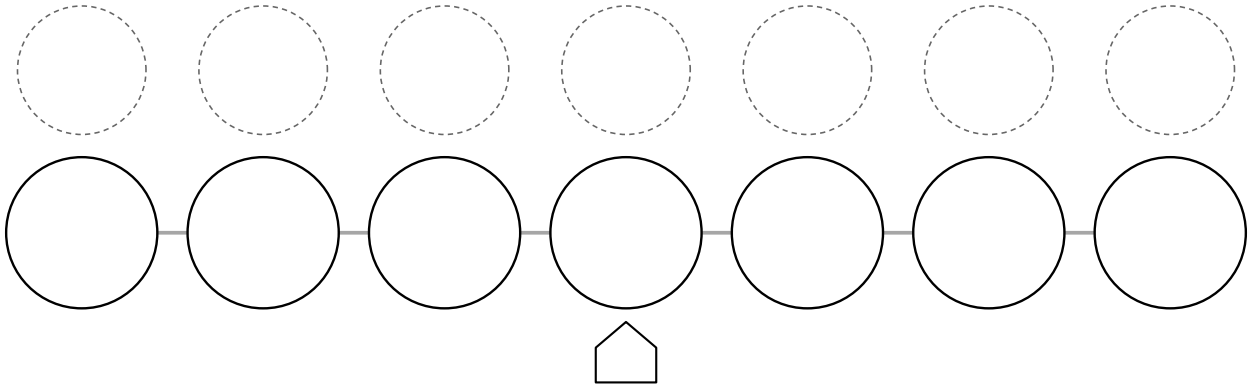
Problem 2: Build each picture with as few moves as you can.



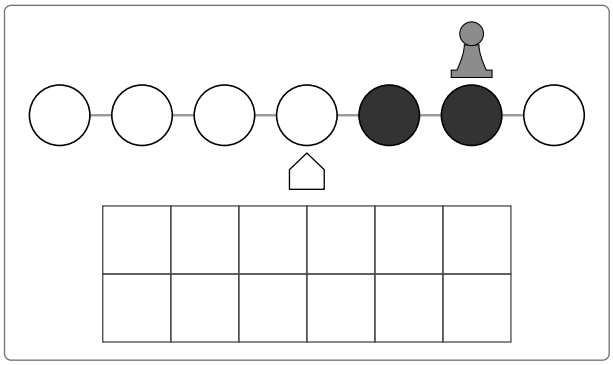


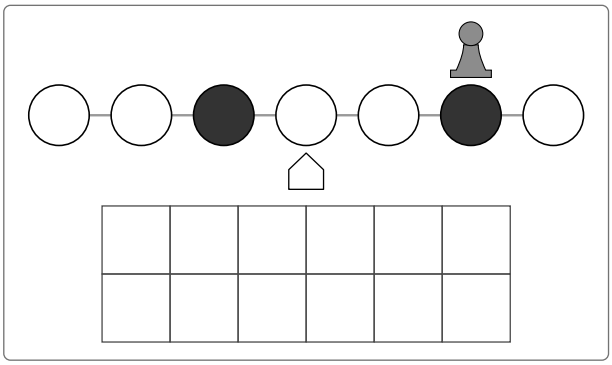


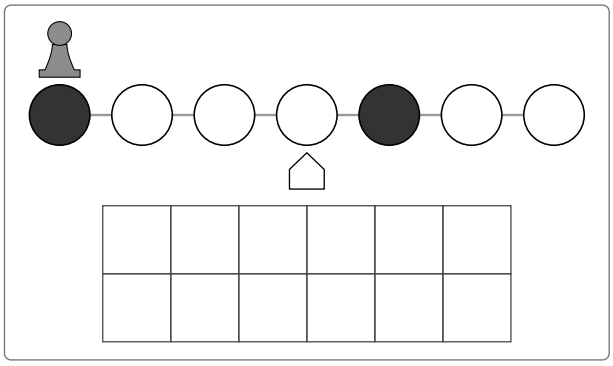


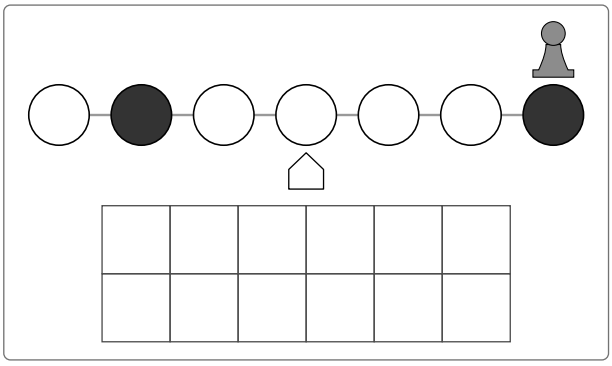


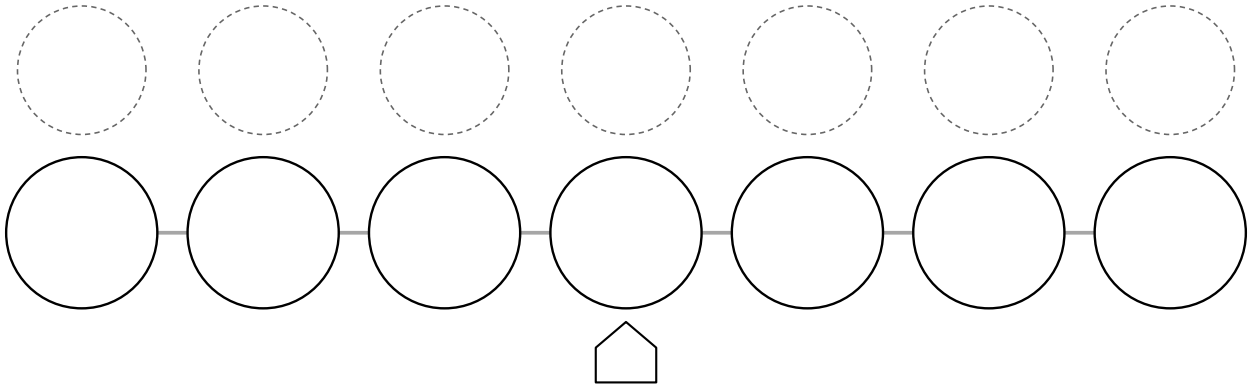
Problem 3: Build each picture with as few moves as you can.



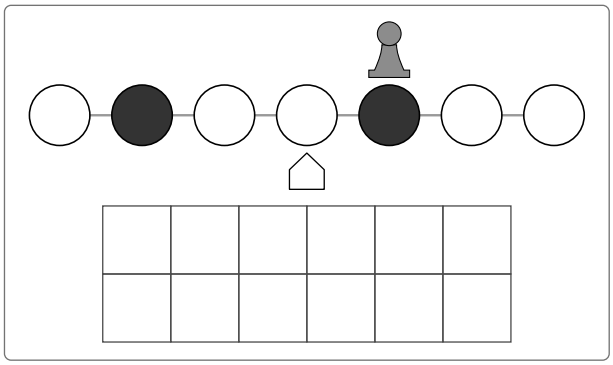


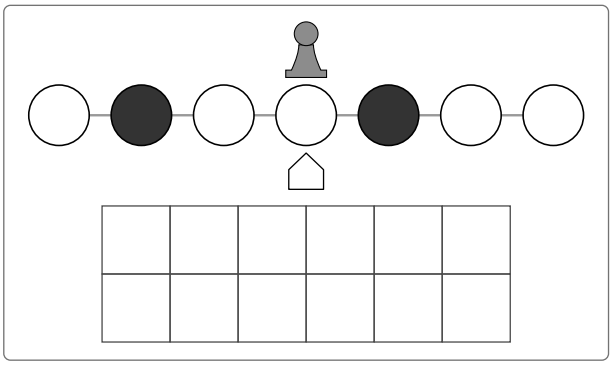


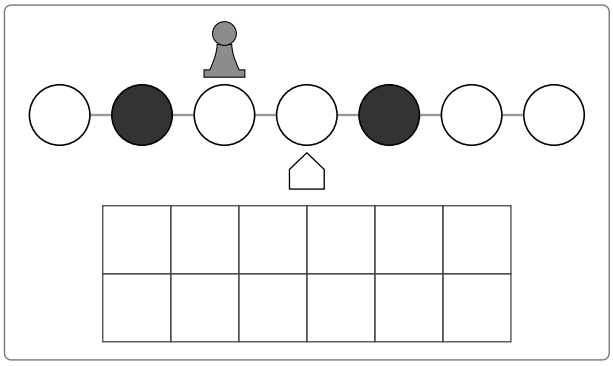


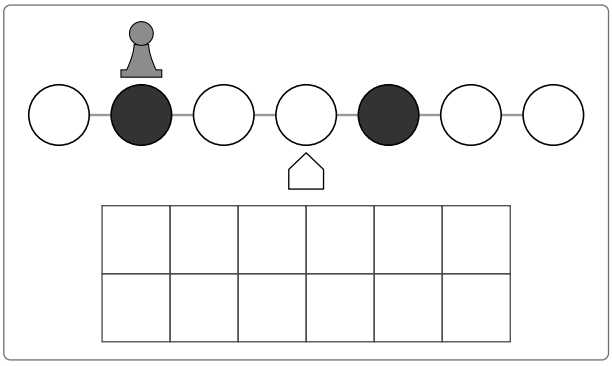


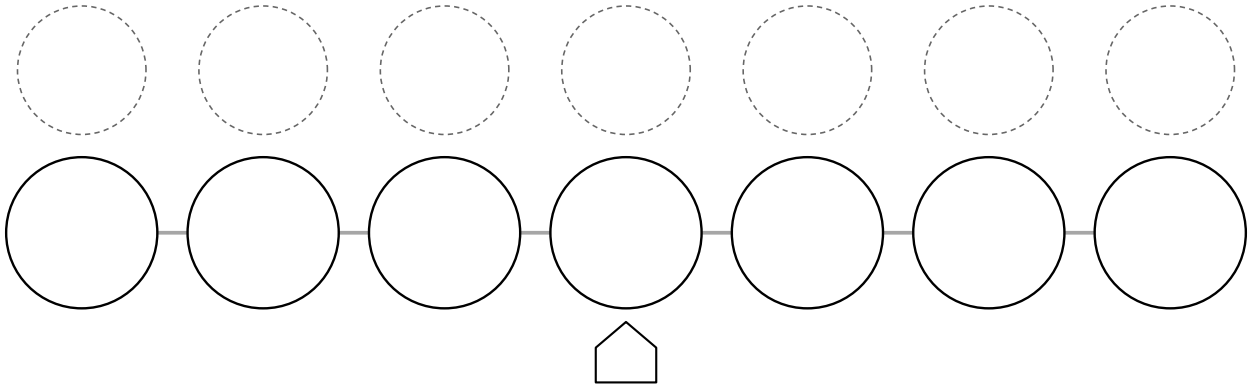
Problem 4: Build each picture with as few moves as you can.



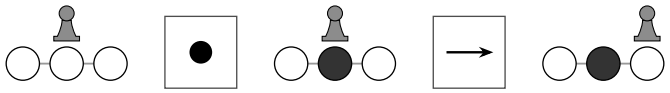


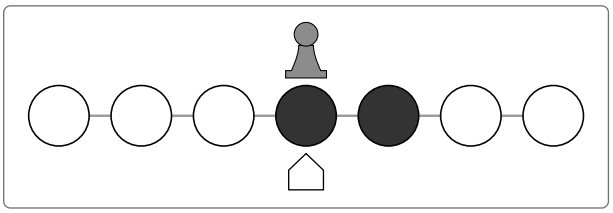


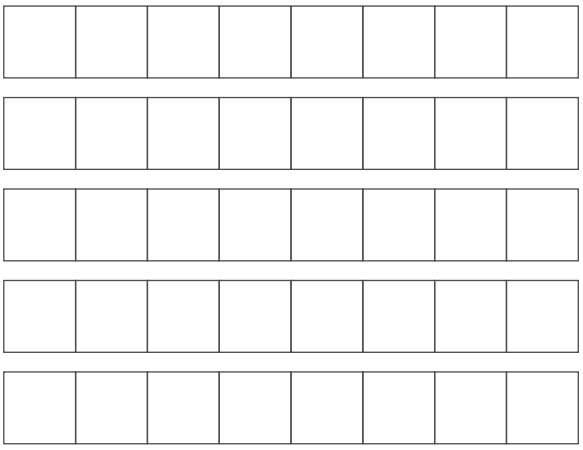


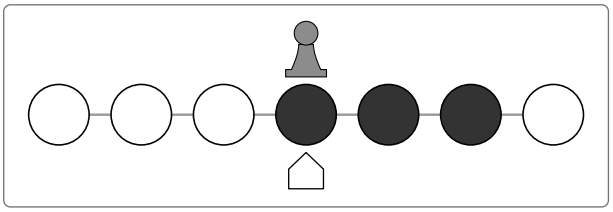


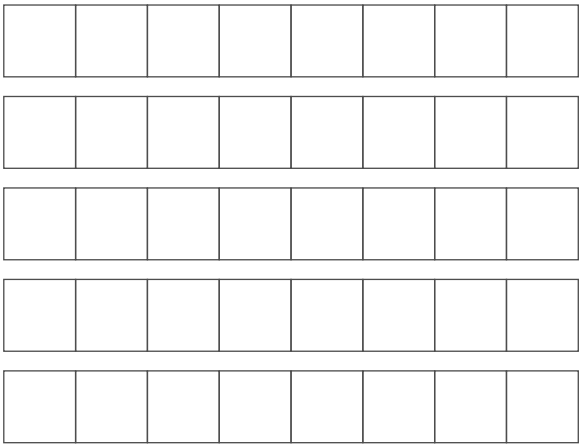
Problem 5: Find every way to build each picture with the fewest moves. Draw each way in a row of squares, with a dot for each flip and an arrow for each step. Explain why there is no quicker way.

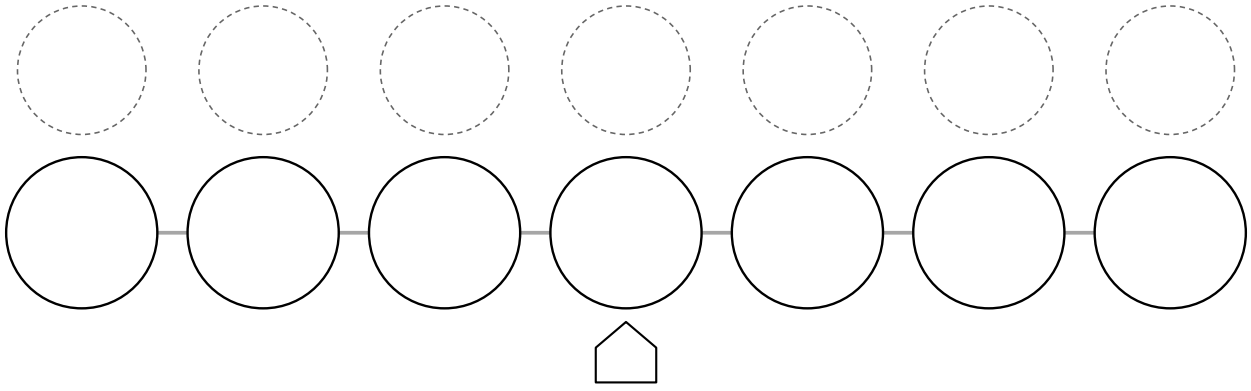




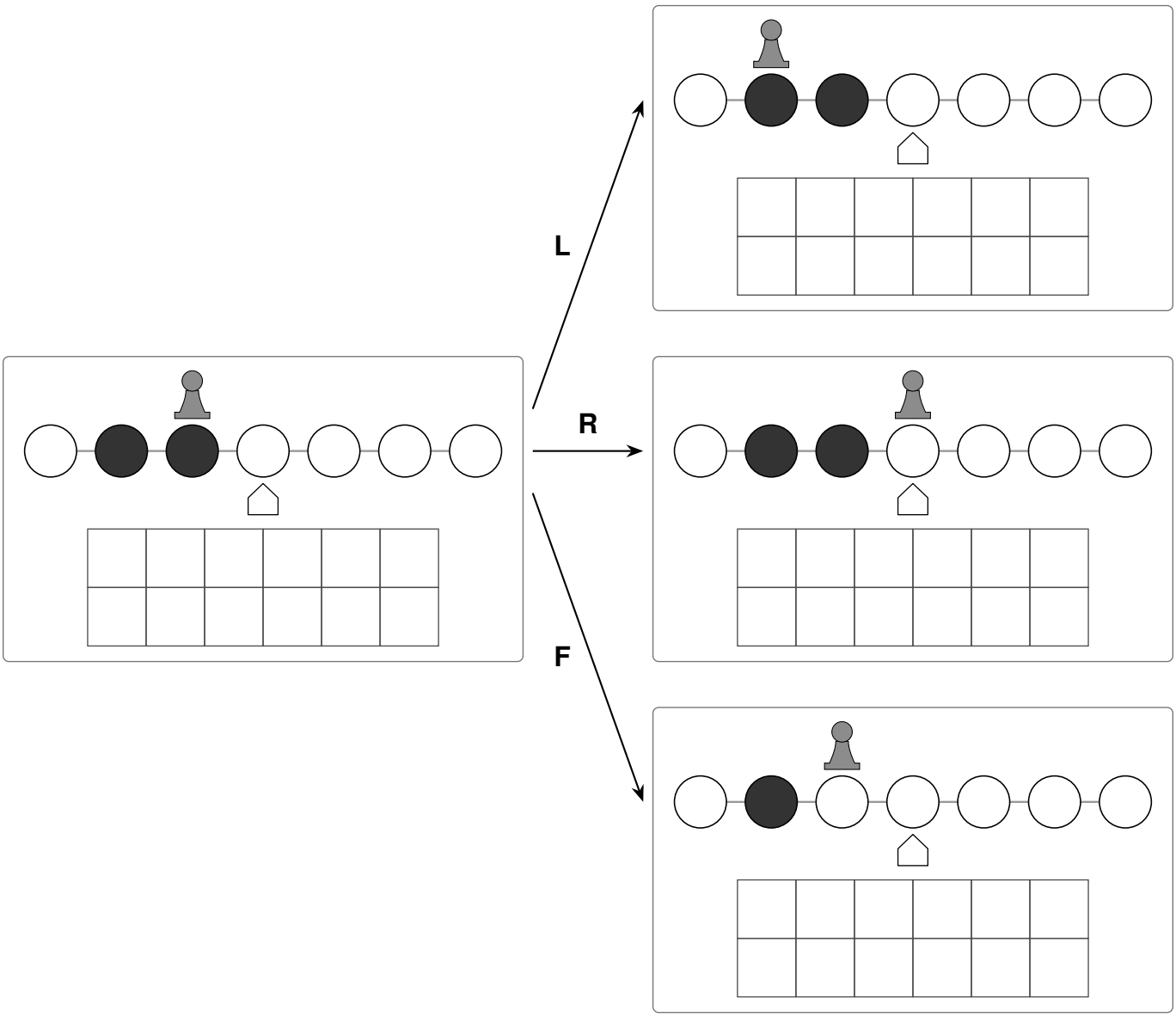


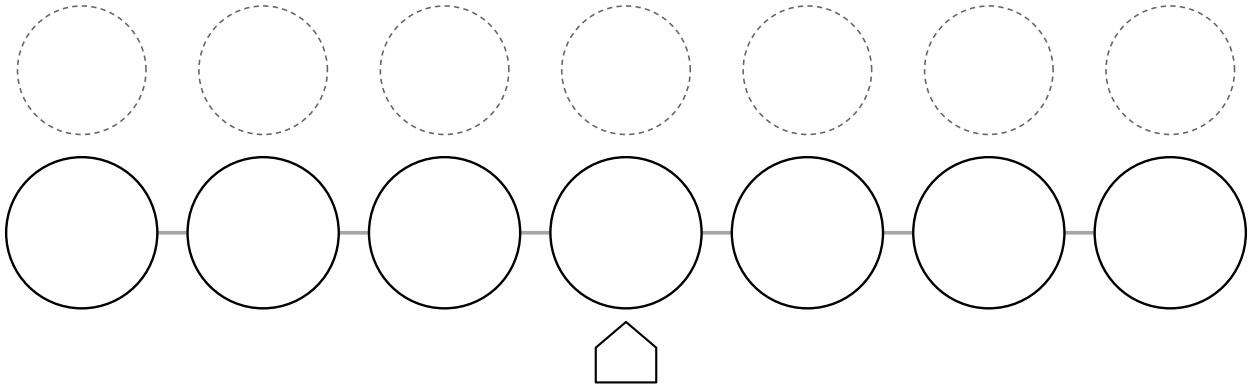




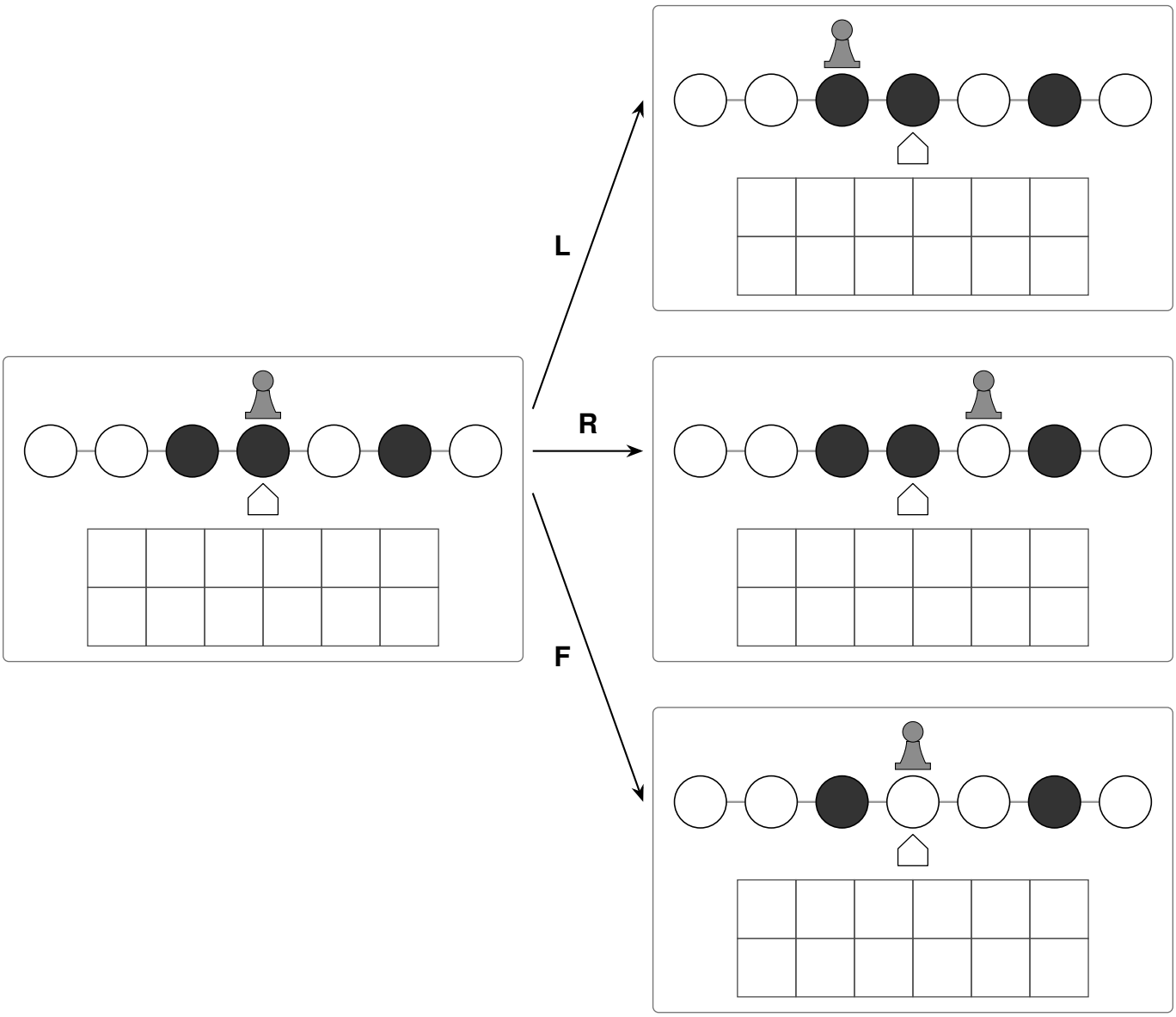


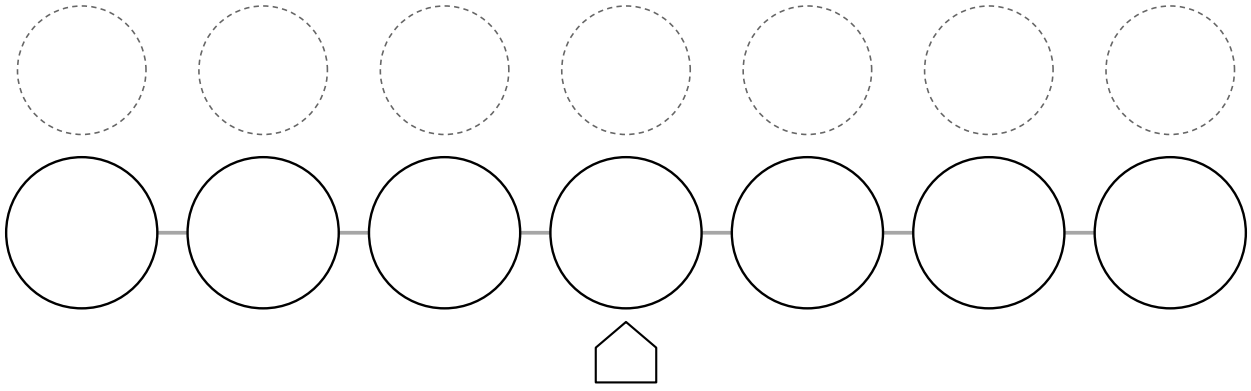
Problem 6: Build each picture, starting with all lamps off, with as few moves as you can. Which one needs the most moves?





Problem 7: Build each picture, starting with all lamps off, with as few moves as you can. Which one needs the most moves?





Problem 8: For each tower, try to build the picture in its row with exactly that many moves. Circle every tower that works.

